1. Project Overview and Stakeholder List

1.1 Full Overview of the Group System

Local Government Units (LGUs) across Camarines Sur rely on real property taxation (RPT) as their primary statutory foundation for local fiscal autonomy under Title II of Republic Act No. 7160 (Local Government Code of 1991). However, municipal tax operations are crippled by severe operational frictions: fragmented departmental record-keeping, manual property valuation prone to human calculation errors, decades-old physical archives dating back to 1949, and an entrenched physical dependency on the Provincial Capitol in Pili for mandatory statutory approvals. To solve these systemic bottlenecks, our research group proposes the Real Property Assessment Tracking and Inter-Office Synchronization System (RPAT-IOSS), an integrated 3-tier web platform designed to eliminate inter-office data silos and compress property assessment and ownership transfer turnaround times from over 7–14 days down to less than 24 hours. The platform comprises three interlocking subsystems:

eRPT Assessment, Valuation & Tax Declaration Engine — Automates statutory base market value appraisal, assessment level formulas, and electronic Tax Declaration (TD), Notice of Assessment (NOA), and Property Appraisal Schedule (PAAS) generation, mitigating calculation errors and multi-day paper preparation.

Inter-Office Treasury Synchronization & Ledger Reconciliation Module — Dynamically synchronizes approved municipal assessment rolls with treasury collection databases in real time, computing statutory RPT dues, early-bird discounts (20%/10%), and delinquency penalties (2%/month) while instantly generating electronic tax clearances.

(Individual Focus): Provincial Capitol Remote Endorsement & Digital Approval Pipeline — Establishes a secure, authenticated digital review, annotation, and approval bridge connecting component municipal assessors directly with the Provincial Assessor's Office at the Provincial Capitol in Pili, eliminating physical road transit and cutting multi-week provincial approval queues.

1.2 Subsystem Focus: Provincial Capitol Remote Endorsement & Digital Approval Pipeline

Under RA 7160 and Camarines Sur provincial administration rules, municipal assessors in component municipalities (such as Pasacao, Canaman, Magarao, and San Fernando) hold only recommending approval authority. Final approval and statutory issuance of the Certificate of Inspection and Survey (CIS) and new Tax Declarations rest exclusively with the Provincial Assessor at the Provincial Capitol in Pili. Currently, municipal personnel must physically travel by road to the Capitol once a week to submit paper dossiers and retrieve endorsements from prior weeks. Busier municipalities wait between 2 to 3 weeks

because the Capitol manually inspects files from 35 component municipalities. This subsystem digitizes this inter-governmental bottleneck: municipal assessors upload certified digital dossiers; provincial examiners inspect and annotate files online; the Provincial Assessor applies cryptographic approval stamps; and finalized endorsements and CIS documents are instantly dispatched back to the LGU in real time.

1.3 Stakeholder List and Primary Interests/Concerns

Stakeholders	Roles and Interest
Municipal Assessors & Staff ***(Assessor's Office: Pasacao, Canaman)***	Issues recommending approvals and manages initial filings. Focuses on instant digital dossier uploads, tracking the Capitol review queue, and eliminating physical road travel to Pili while addressing road hazards, long wait-time visibilities, and citizen complaints.
Provincial Assessor & Approvers ***(Provincial Capitol: Pili)***	Conducts statutory compliance audits and rapid verification across 35 municipalities in Camarines Sur. Focuses on reducing backlog, preventing physical document decay, and eliminating incomplete paper folders that require return.
Provincial Review Examiners ***(Technical Review Unit: Pili)***	Performs online document inspection, digital annotation of boundary/rate errors, and structured audit checklists. Replaces tedious physical paper cross-referencing, title marking, and slow phone follow-ups.
Municipal Treasurers & Cashiers ***(Municipal Treasury Office)***	Requires real-time access to approved assessments to compute tax dues, apply discounts, and issue tax clearances, resolving tax collection delays caused by pending provincial approvals.
Property Owners & Taxpayers ***(Citizens, Buyers, Heirs)***	Submits requests for land transfers and assessments. Needs rapid transfer processing, transparent milestone tracking, and fewer counter visits to avoid idle waiting times and repeated follow-up trips.

System Administrators & IT Staff ***(LGU & Provincial IT Units)***	Evaluates academic rigor, distinct modular contributions per student, and measurable latency reduction under ISO 25010 standards, ensuring project uniqueness from adjacent capstone works.

2. Information-Gathering Methods Used and Summary of Findings

2.1 Techniques Employed

To establish empirical grounding, four complementary information-gathering techniques were executed across target LGUs and academic consultations:

Semi-Structured Field Interviews: Conducted structured on-site interviews with key informants: the Municipal Assessor Office Head and 4 staff members at Canaman LGU, Canaman Municipal Treasury personnel, and Tax Mapper I (John Francisco) alongside assessment personnel at Pasacao LGU. Investigated transaction frequencies, documentary requirements, calculation practices, and provincial routing.

Field Observation & Workflow Walkthrough: Directly observed citizen intake, counter queue dynamics, spreadsheet lookup methods, physical archive folder retrieval, and the manual bundling of paper dossiers destined for the weekly car run to the Provincial Capitol in Pili.

Faculty & Academic Consultations: Participated in technical consultation sessions with research advisers (Sir Chin, Sir Arreb) and CSACIS capstone faculty to establish project boundaries, validate the provincial endorsement bottleneck, and ensure distinct project uniqueness.

2.2 Summary of Key Findings

The Provincial Courier Bottleneck: Municipal assessors in Camarines Sur do not have final approval authority. Endorsements must be approved by the Provincial Capitol in Pili. Municipal staff courier documents by vehicle once a week (Thursdays/Fridays). While Canaman takes ~1 week due to lower volume, busier municipalities wait 2 to 3 weeks because the Capitol services 35 component municipalities.

The Pasacao Computerization Gap: While Pasacao utilizes localized software (eRPT, upgraded 2024) that reduced routine certifications to 5–15 minutes, property transfers still take up to 7 working days. The system remains completely internal to assessor workstations, lacking citizen visibility and dynamic integration with the Provincial Capitol and Municipal Treasury.

Departmental Data Silos: Assessor and Treasurer offices operate in complete isolation. Assessors generate assessment rolls on paper/Excel, while the Treasury manually tracks payments. Citizens must physically shuttle receipts between windows to verify tax status and obtain clearances.

Archival Search Latency: In manual LGUs like Canaman, active records take 10–30 minutes to retrieve, but retrieving historical records dating back to 1949

takes 2 to 3 full business days in physical archive vaults, with severe risks of document decay.

3. Functional Requirements (Prioritized)

Priority	Requirement Title	Detailed Functional Specification
Must Have	Municipal Endorsement Docket Bundling	The system MUST allow authorized Municipal Assessor staff to bundle and upload digital endorsement packets containing appraisal sheets, scanned deeds of conveyance, land survey sketches, and certified ROD titles in encrypted PDF/PNG formats (<= 25MB).
Must Have	Provincial Remote Review Workspace	The system MUST provide Provincial Review Examiners with a web interface featuring a dual-pane document inspection viewer, zoom controls, and an interactive statutory compliance checklist to audit submitted municipal dockets.
Must Have	Digital Approval & Cryptographic Stamping	The system MUST enable the Provincial Assessor to execute official approvals online, automatically affixing a cryptographic SHA-256 signature stamp, unique verification token, and digital approval watermark.
Must Have	Structured Deficiency Flagging & Return	The system MUST allow provincial examiners to flag specific documentary or valuation deficiencies (e.g., missing notarization, boundary dispute, rate mismatch) and return the docket with itemized corrective notes and instant municipal notification.
Should Have	Role-Based Access Control (RBAC)	The system SHOULD enforce multi-tier role permissions separating Municipal Assessor Clerks, Municipal Assessor Heads, Provincial Examiners, the Provincial Assessor, and System Auditors.

Should Have	Automated CIS Generation	The system SHOULD automatically generate a standardized, printable Certificate of Inspection and Survey (CIS) with an embedded verification QR code immediately upon provincial approval.
Should Have	Real-Time Bidirectional Dispatch	The system SHOULD dispatch approved dockets or revision notices back to municipal assessor and treasury databases in under 1.0 second, updating milestone status across all dashboards.
Should Have	Immutable Transaction Audit Logging	The system SHOULD maintain an append-only audit log recording every docket creation, view, annotation, approval, rejection, user ID, IP address, and timestamp.
Could Have	Automated SMV Variance Checking	The system COULD cross-reference municipal appraisal inputs against the provincial Schedule of Base Market Values table, automatically alerting examiners to discrepancies in land classifications or unit rates.
Could Have	Batch Endorsement Approval	The system COULD permit the Provincial Assessor to select multiple pre-audited dockets and apply digital approvals in a single batch transaction during peak volume cycles.
Won't Have	Provincial Mainframe Direct Sync	The system WON'T execute live queries directly into the Provincial Capitol's internal on-premise mainframe during this release due to provincial cybersecurity and firewall policies.
Won't Have	Direct Citizen Filing to Capitol	The system WON'T permit individual citizens to submit applications directly to the Provincial Capitol; all dockets must strictly originate from and be endorsed by the component Municipal Assessor.

4. Non-Functional Requirements (Prioritized & Measurable)

Non-functional requirements specify quality attributes and security constraints framed around the ISO/IEC 25010 Software Quality Standard:

Quality Dimension	Measurable Specification Target	MoSCoW
Security & Privacy	All transmissions must enforce TLS 1.3 encryption. All stored property files and personal identifiers must be encrypted using AES-256 at rest. Strict compliance with RA 10173 (Data Privacy Act). Idle user sessions must terminate automatically after 15 minutes.	**Must Have**
Performance Efficiency	Web dashboard pages must render in under 2.0 seconds over a 10 Mbps connection. Upload of dockets up to 25MB must complete within 5.0 seconds. Endorsement status updates must synchronize across municipal and provincial views within 1.0 second.	**Must Have**
Data Integrity	Digital approval stamps must maintain cryptographic hash verification (SHA-256). Any post-approval tampering with a dossier must immediately invalidate its QR token and trigger an administrative security alert.	**Must Have**
Usability & Learnability	The UI must achieve a System Usability Scale (SUS) score >= 80/100 and an ISO/IEC 25010 Usability rating >= 4.0/5.0. Municipal and provincial personnel must achieve operational competency with under 30 minutes of orientation.	**Should Have**
Reliability & Availability	The system must maintain >= 99.5% uptime during government operating hours (8:00 AM – 5:00 PM, Monday through Friday). Automated incremental database backups must occur every 24 hours.	**Should Have**

Compatibility	The application must be fully responsive across desktop, laptop, and tablet displays (resolutions 1024x768 to 1920x1080) and support modern web browsers (Chrome, Edge, Firefox, Safari) without proprietary plugins.	**Should Have**
Maintainability	Backend services must be structured as modular RESTful APIs with OpenAPI/Swagger documentation, enabling future microservice integration with national DICT eGov or eRPTS platforms.	**Could Have**

5. Key Use Cases or User Stories (High Priority)

The operational workflows of the subsystem are specified through three high-priority use cases and five prioritized user stories:

5.1 Use Case 1: Municipal Assessor Bundles and Submits Remote Endorsement (UC-01)

Actor: Municipal Assessor / Assessment Appraiser

Goal: Package and transmit an evaluated property assessment dossier to the Provincial Capitol digitally without physical road transit.

Preconditions: Municipal assessor has completed local appraisal and logged in with valid credentials.

Main Steps: 1. Assessor navigates to 'Capitol Endorsement' module. 2. Selects pending transaction (e.g., Transfer of Assessment). 3. Uploads certified digital documents (scanned Deed of Sale, ROD Title, Field Appraisal Form, Lot Sketch <= 25MB). 4. System validates file integrity and generates SHA-256 hash. 5. Assessor clicks 'Submit Recommending Endorsement'. 6. System marks status 'Under Provincial Review' and queues docket in Capitol inbox.

Postconditions: Endorsement packet is queued in Provincial Capitol inbox with an immutable timestamp, visible on municipal tracking dashboards.

5.2 Use Case 2: Provincial Review Examiner Inspects and Annotates Docket (UC-02

Actor: Provincial Technical Review Examiner

Goal: Perform structured compliance auditing and digital annotation on submitted municipal dockets.

Preconditions: Docket is in 'Under Provincial Review' status; examiner has active Capitol reviewer credentials.

Main Steps: 1. Examiner selects pending municipal docket from provincial queue. 2. System displays split-screen view: digitized PDF dossier viewer on left, statutory compliance checklist on right. 3. Examiner audits property boundaries, land classification, and conveyance deeds. 4. Marks compliance checkboxes. If deficiency is found: examiner highlights error, adds structured corrective remark, and clicks 'Return for Revision' (status changes to

'Deficiency Flagged' with instant notification). 5. If compliant: examiner clicks 'Endorse for Final Approval'.

Postconditions: Docket is approved for signature and forwarded to the Provincial Assessor's final signing queue.

5.3 Use Case 3: Provincial Assessor Applies Digital Approval and Issues CIS (UC-03)

Actor: Provincial Assessor / Authorized Approving Official

Goal: Affix cryptographic digital approval, generate official CIS, and dispatch finalized endorsement to the LGU.

Preconditions: Docket has been audited and recommended for approval by the Provincial Review Examiner.

Main Steps: 1. Provincial Assessor opens recommended docket and inspects examiner's compliance summary. 2. Clicks 'Authorize & Execute Digital Approval'. 3. System prompts for authentication PIN/2FA signing token. 4. System affixes cryptographic signature, generates verification QR code, and compiles official Certificate of Inspection and Survey (CIS). 5. System updates transaction status to 'Provincially Approved'. 6. System pushes approved digital package back to Municipal Assessor and Treasury databases in real time.

Postconditions: Property transaction is officially approved. Municipal Assessor can print the final Tax Declaration, and Treasury can issue real property tax bills.

5.4 Prioritized User Stories

User Story 1 (High Priority): "As a Municipal Assessor, I want to bundle and transmit assessment dossiers online to the Provincial Capitol so that our staff eliminates weekly car trips to Pili and avoids 2-to-3-week courier delays."

User Story 2 (High Priority): "As a Provincial Review Examiner, I want to inspect submitted dockets and annotate discrepancies within a structured web interface so that incomplete files are caught immediately without physical paper damage."

User Story 3 (High Priority): "As the Provincial Assessor, I want to execute legally binding cryptographic approvals and generate verified CIS certificates online so that component municipalities receive finalized approvals instantaneously."

User Story 4 (High Priority): "As a Municipal Assessment Clerk, I want to monitor live Capitol review milestone statuses online so that I can provide instant, transparent progress updates to taxpayers inquiring at the counter."

User Story 5 (Medium Priority): "As a Municipal Treasurer, I want to receive approved assessment records automatically via system synchronization so that accurate tax bills and clearances can be generated without manual delays."

6. Assumptions, Constraints, and Open Issues

6.1 System Assumptions

Broadband Connectivity: Municipal Assessor offices (Pasacao, Canaman) and the Provincial Capitol in Pili maintain baseline broadband connections (minimum 10 Mbps).

Hardware Readiness: Participating offices possess standard desktop PCs running modern web browsers capable of rendering PDF documents and responsive dashboards.

Administrative Willingness: Provincial Assessor leadership in Pili will permit designated personnel to participate in controlled pilot trials and evaluation surveys.

Legal Recognition: Electronic approvals, cryptographic hashes, and digital stamps produced by the system are recognized as valid administrative acts under the Electronic Commerce Act of 2000 (RA 8792).

6.2 System Constraints

Technical Constraints: Direct read/write access to internal mainframe servers of the Provincial Capitol is prohibited by provincial cybersecurity policies. The pipeline must operate as an independent, authenticated web portal.

Policy & Statutory Constraints: Under Section 202 of RA 7160, component municipalities cannot grant final assessment approvals. The system must strictly preserve the statutory two-tiered hierarchy (Municipal Recommending -> Provincial Final Approval).

Data Privacy Constraints: Must strictly conform to the Data Privacy Act of 2012 (RA 10173) regarding land titles, taxpayer identification, property valuations, and financial figures.

Budgetary Constraints: Software architecture must utilize open-source frameworks (Python FastAPI/Node.js, React, PostgreSQL) to avoid software licensing burdens for resource-constrained LGUs.

Time Constraints: Software development, simulation trials, pilot testing, and thesis manuscript defense must be fully accomplished within the 6-month capstone period (July to December 2026).

6.3 Open Issues and Questions

PNPKI Integration vs. Platform Tokens: Determining whether the Provincial Assessor will require integration with the DICT Philippine National Public Key Infrastructure (PNPKI) digital certificate authority, or if platform-level SHA-256 tokens and secure QR stamps are sufficient for initial deployment.

Physical Document Retention Policy: Establishing whether provincial examiners will require periodic post-audit physical inspections of original notarized deeds of conveyance, even when digital dossiers have been approved online.

7. Attachment of Data Gathering Evidences

Target Agency / Location	Informants Engaged	Key Evidentiary Data Gathered
Municipal Assessor's Office, Canaman, Camarines Sur	4 Staff Members + 1 Office Head (5 personnel total)	Documented manual ledger lookups, 1949 physical archive vaults (2-3 days search time), conveyance requirements, and weekly Thursday/Friday courier runs to Capitol in Pili.
Municipal Treasurer's Office, Canaman, Camarines Sur	Municipal Treasury Officer & Revenue Staff	Confirmed disconnected assessor rolls, real property tax billing workflows, 20% advance discount (Dec), 10% Q1 discount, and 2%/month delinquency penalties.
Municipal Assessor's Office, Pasacao, Camarines Sur	John Francisco (Tax Mapper I) & Assessor Office Staff	Benchmarked eRPT software (upgraded 2024), 50–100 daily certification volume, 7-day transfer delays, and complete absence of citizen tracking.
UNC CSACIS Faculty Office	Sir Chin & Sir Arreb (Research Advisers)	Advised on scoping to local municipal workflows, avoiding overlap with adjacent GIS mapping project, and framing the study around an empirical Needs Assessment.
UNC CSACIS Capstone Laboratory	Capstone Research Professor	Analyzed DICT eRPT open-source limitations, municipal server constraints, legacy data migration (1949–2000), and microservice integration architecture.

7.2 Standardized Interview Questions Administered

Question 1: What are the most frequent real property assessment transactions handled daily and monthly in your office?

Question 2: What specific documentary requirements are mandatory for a Transfer of Assessment, and how are non-owner representatives validated?

Question 3: What is the step-by-step workflow for property appraisal, and how are base market values and assessed values calculated?

Question 4: How are property records stored, indexed, and retrieved? How much time is spent retrieving historical records (dating back to 1949)?

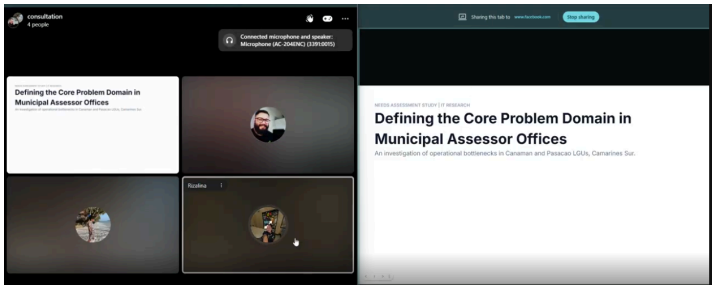
Question 5: How does your office coordinate with the Municipal Treasurer's Office, and how are assessment rolls and tax payment records reconciled?

Question 6: What is the exact process and turnaround time for securing endorsements and approvals from the Provincial Capitol in Pili?

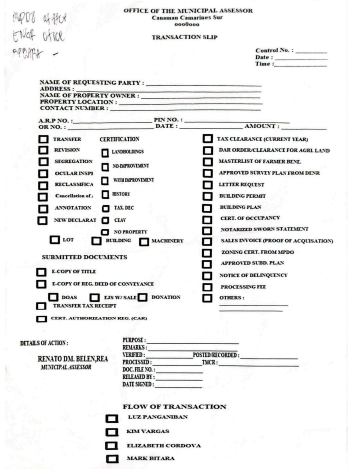
Question 7: What software tools (e.g., eRPT, Excel) are currently deployed, what are their main technical limitations, and what features would optimize operations?







Forms



OFFICE OF THE MUNICIPAL ASSESSOR
Canaman Camarines Sur
TRANSACTION SLIP

Control No. :
Date :
Time :

NAME OF REQUESTING PARTY :
ADDRESS :
NAME OF PROPERTY OWNER :
PROPERTY LOCATION :
CONTACT NUMBER :

A.R.P NO. : PIN NO. :
OR NO. : DATE : AMOUNT :

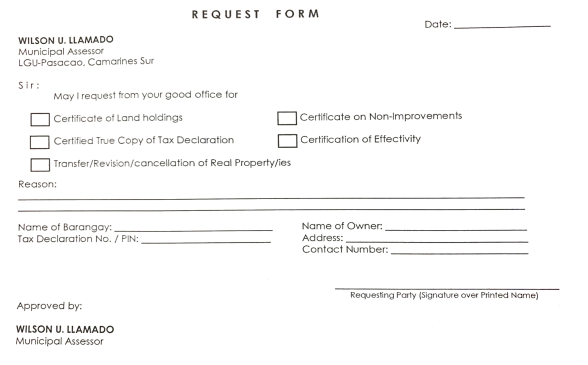
TRANSFER
REVISION
SEGREGATION
OCULAR INSP.
RECLASSIFICA
Cancellation of:
ANNOTATION
NEW DECLARAT
CERTIFICATION
LAND/BLDGS
NO IMPROVEMENT
WITH IMPROVEMENT
HISTORY
TAX DEC
CLAY
NO PROPERTY
LOT
BUILDING
MACHINERY
TAX CLEARANCE (CURRENT YEAR)
DAR ORDER/CLEARANCE FOR AGRI. LAND
MASTERLIST OF FARMER BENE.
APPROVED SURVEY PLAN FROM DENR
LETTER REQUEST
BUILDING PERMIT
BUILDING PLAN
CERT. OF OCCUPANCY
NOTARIZED SWORN STATEMENT
SALES INVOICE (PROOF OF ACQUISATION)
ZONING CERT. FROM MPDO
APPROVED SUBD. PLAN
NOTICE OF DELINQUENCY
PROCESSING FEE
OTHERS :

SUBMITTED DOCUMENTS
E-COPY OF TITLE
E-COPY OF REG. DEED OF CONVEYANCE
DOAS
EJS W/ SALE
DONATION
TRANSFER TAX RECEIPT
CERT. AUTHORIZATION REG. (CAR)

DETAILS OF ACTION :
PURPOSE :
REMARKS :
VERIFIED : POSTED/RECORDED :
PROCESSED : TAGS :
DOC. FILE NO. :
RELEASED BY :
DATE SIGNED :

RENATO DM. BELEN,REA
MUNICIPAL ASSESSOR

FLOW OF TRANSACTION
LUZ PANGANIBAN
KIM VARGAS
ELIZABETH CORDOVA
MARK BITARA



REQUEST FORM

Date: _____

WILSON U. LLAMADO
Municipal Assessor
LGU-Pasacao, Camarines Sur

Sir:

May I request from your good office for

☐ Certificate of Land holdings
☐ Certificate on Non-Improvements
☐ Certified True Copy of Tax Declaration
☐ Certification of Effectivity
☐ Transfer/Revision/cancellation of Real Property/ies

Reason:
_____
_____

Name of Barangay: _____
Tax Declaration No. / PIN: _____

Name of Owner: _____
Address: _____
Contact Number: _____

_____
Requesting Party (Signature over Printed Name)

Approved by:

WILSON U. LLAMADO
Municipal Assessor